

Then we decompose the last integral and estimate

$$\begin{aligned} & \sum_{j=0}^{\infty} \int_{2^{-j-1}\delta < |x-y| \leq 2^{-j}\delta} \frac{1}{|x-y|^{d-1}} |W^{\frac{1}{p}}(x)W^{-\frac{1}{p}}(y)W^{\frac{1}{p}}(y)\vec{f}(y)| dy \\ & \lesssim \delta \sum_{j=0}^{\infty} 2^{-j} \int_{|x-y| \leq 2^{-j}\delta} |W^{\frac{1}{p}}(x)W^{-\frac{1}{p}}(y)W^{\frac{1}{p}}(y)\vec{f}(y)| dy \\ & \lesssim \delta M_W(W^{\frac{1}{p}}\vec{f})(x), \end{aligned}$$

where M_W is the matrix-weighted maximal function (4). Due to the boundedness of the matrix-weighted maximal function we get

$$\|[b, T_{\Omega}]\vec{f} - [b, T_{\Omega, \delta}]\vec{f}\|_{L^p(W)} \lesssim_{b, \Omega} \delta \|M_W(W^{\frac{1}{p}}\vec{f})\|_{L^p(\mathbb{R}^d)} \lesssim_W \delta \|\vec{f}\|_{L^p(W)}$$

Since δ can be arbitrarily small, it suffices to show that $[b, T_{\Omega, \delta}]: L^p(W) \rightarrow L^p(W)$ is compact. We note that the truncated kernel satisfies

$$(5) \quad |K_{\Omega, \delta}(x, y) - K_{\Omega, \delta}(x', y)| \lesssim \frac{|x - x'|}{|x - y|^{d+1}}, \quad 2|x - x'| \leq |x - y|.$$

By [17, Corollary 6.3] we have that $[b, T_{\Omega}]$ is $L^p(W) \rightarrow L^p(W)$ bounded, and hence

$$\begin{aligned} \|[b, T_{\Omega, \delta}]\vec{f}\|_{L^p(W)} & \leq \|[b, T_{\Omega, \delta}]\vec{f} - [b, T_{\Omega}]\vec{f}\|_{L^p(W)} + \|[b, T_{\Omega}]\vec{f}\|_{L^p(W)} \\ & \lesssim_{b, W, \Omega} \|\vec{f}\|_{L^p(W)} < \infty, \end{aligned}$$

which takes care of the first part of Theorem 5.1.

Considering the second part, we suppose that Q is a cube centered at the origin such that $\text{supp } b \subset Q$. Then for x sufficiently far away from the origin we have

$$\begin{aligned} |W^{\frac{1}{p}}(x)[b, T_{\Omega, \delta}]\vec{f}(x)|^p & = \left| W^{\frac{1}{p}}(x) \int_{\mathbb{R}^d} b(y) K_{\Omega, \delta}(x, y) \vec{f}(y) dy \right|^p \\ & \lesssim_{b, \Omega} |W^{\frac{1}{p}}(x)|_{op}^p \frac{1}{|x|^{dp}} \left(\int_Q |\vec{f}(y)| dy \right)^p \\ & \leq |W(x)|_{op} \frac{1}{|x|^{dp}} \left(\int_Q |\vec{f}(y)| dy \right)^p, \end{aligned}$$

where the last inequality is true due to the Cordes inequality. Moreover, an application Hölder's inequality and another application of the Cordes inequality yields

$$\int_Q |\vec{f}(y)| dy \leq \int_Q |W^{-\frac{1}{p}}(y)|_{op} |W^{\frac{1}{p}}(y)\vec{f}(y)| dy \leq \left(\int_Q |W^{-\frac{p'}{p}}(y)|_{op} dy \right)^{\frac{1}{p'}} \|\vec{f}\|_{L^p(W)}.$$